ABSTRACT

A method and system for establishing a communication link between two points on or near the Earth's surface by using non-cooperative orbiting objects, such as satellites or other space debris, as passive reflectors for a directed radio signal. The system does not require any active participation or modification of the orbiting objects. A ground-based system calculates the future position of a suitable object that will momentarily align to reflect a signal from a transmitter to a receiver. By precisely timing the transmission to coincide with the object's passage through the reflection point, a transient communication link is created. The invention provides a means for long-distance, over-the-horizon communication that is difficult to in- tercept and does not rely on traditional geostationary communication satellites or terrestrial infrastructure. The system can operate with either movable, directable antennas or an array of fixed-orientation antennas.